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United States Patent	12417777
Kind Code	B2
Date of Patent	September 16, 2025
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Information processing device and method for outputting a target sound signal from a mixed sound signal

Abstract

An information processing device includes an acquisition unit that acquires sound source position information and so forth, a sound feature value extraction unit that extracts a plurality of sound feature values based on the mixed sound signal, an emphasis unit that emphasizes a sound feature value in a target sound direction based on the sound source position information, an estimation unit that estimates the target sound direction based on the sound source position information and so forth, a mask feature value extraction unit that extracts a mask feature value based on the estimated target sound direction and so forth, a generation unit that generates a target sound direction emphasis sound signal and generates a target sound direction masking sound signal, and a target sound signal output unit that outputs a target sound signal by using the target sound direction emphasis sound signal and so forth.

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Appl. No.:	18/239289
Filed:	August 29, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230419980 A1	Dec. 28, 2023

Related U.S. Application Data

continuation parent-doc WO PCT/JP2021/014790 20210407 PENDING child-doc US 18239289

Publication Classification

Int. Cl.: G10L21/0308 (20130101); G10L21/0216 (20130101); G10L25/18 (20130101); G10L25/21 (20130101)

U.S. Cl.:

CPC G10L21/0308 (20130101); G10L21/0216 (20130101); G10L25/18 (20130101); G10L25/21 (20130101);

Field of Classification Search

CPC: G10L (21/0308); G10L (21/0216); G10L (25/18); G10L (25/21); G10L (25/51); G10L (21/0264); G10L (2021/02166)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of International Application No. PCT/JP2021/014790 having an international filing date of Apr. 7, 2021.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present disclosure relates to an information processing device, and an output method.

2. Description of the Related Art

(2) Speeches are mixed together when a plurality of speakers speak at the same time. There are cases where it is desired to extract the speech of a target speaker from the mixed speech. When extracting the speech of the target speaker, a method for suppressing noise can be considered, for example. Here, a method for suppressing noise has been proposed (see Patent Reference 1). Patent Reference 1: Japanese Patent Application Publication No. 2010-239424 Patent Reference 2: WO 2016/143125 Non-patent Reference 1: Yi Luo, Nima Mesgarani, "Conv-TasNet: Surpassing Ideal Time-Frequency Magnitude Masking for Speech Separation", 2019 Non-patent Reference 2: Ashish Vaswani et al., "Attention Is All You Need", in Proc. NIPS, 2017

(3) Additionally, when an angle between a direction of incidence of a target sound (e.g., speech of the target speaker) upon a microphone and a direction of incidence of a masking sound (e.g., speech of a disturbing speaker) upon the microphone is small, there are cases where it is difficult for a device to output a target sound signal as a signal representing the target sound even by use of the above-described technology.

SUMMARY OF THE INVENTION

(4) An object of the present disclosure is to output the target sound signal.

(5) An information processing device according to an aspect of the present disclosure is provided. The information processing device includes an acquisition unit that acquires sound source position information as position information on a sound source of a target sound, a mixed sound signal as a signal representing a mixed sound including the target sound and a masking sound, and a learned model, a sound feature value extraction unit that extracts a plurality of sound feature values based on the mixed sound signal, an emphasis unit that emphasizes a sound feature value in a target sound direction as a direction of the target sound among the plurality of sound feature values based on the sound source position information, an estimation unit that estimates the target sound

direction based on the plurality of sound feature values and the sound source position information, a mask feature value extraction unit that extracts a mask feature value, as a feature value in a state in which a feature value in the target sound direction is masked, based on the estimated target sound direction and the plurality of sound feature values, a generation unit that generates a target sound direction emphasis sound signal, as a sound signal in which the target sound direction is emphasized, based on the emphasized sound feature value and generates a target sound direction masking sound signal, as a sound signal in which the target sound direction is masked, based on the mask feature value, and a target sound signal output unit that outputs a target sound signal as a signal representing the target sound by using the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model.

(6) According to the present disclosure, the target sound signal can be outputted.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure will become more fully understood from the detailed description given hereinbelow and the accompanying drawings which are given by way of illustration only, and thus are not limitative of the present disclosure, and wherein:
- (2) FIG. 1 is a diagram showing an example of a target sound signal output system in a first embodiment;
- (3) FIG. 2 is a diagram showing hardware included in an information processing device in the first embodiment;
- (4) FIG. 3 is a block diagram showing functions of the information processing device in the first embodiment;
- (5) FIG. 4 is a diagram showing a configuration example of a learned model in the first embodiment;
- (6) FIG. 5 is a flowchart showing an example of a process executed by the information processing device in the first embodiment;
- (7) FIG. 6 is a block diagram showing functions of a learning device in the first embodiment;
- (8) FIG. 7 is a flowchart showing an example of a process executed by the learning device in the first embodiment;
- (9) FIG. 8 is a block diagram showing functions of an information processing device in a second embodiment;
- (10) FIG. 9 is a flowchart showing an example of a process executed by the information processing device in the second embodiment;
- (11) FIG. 10 is a block diagram showing functions of an information processing device in a third embodiment;
- (12) FIG. 11 is a flowchart showing an example of a process executed by the information processing device in the third embodiment;
- (13) FIG. 12 is a block diagram showing functions of an information processing device in a fourth embodiment; and
- (14) FIG. 13 is a flowchart showing an example of a process executed by the information processing device in the fourth embodiment.

DETAILED DESCRIPTION OF THE INVENTION

(15) Embodiments will be described below with reference to the drawings. The following embodiments are just examples and a variety of modifications are possible within the scope of the present disclosure.

First Embodiment

- (16) FIG. 1 is a diagram showing an example of a target sound signal output system in a first

embodiment. The target sound signal output system includes an information processing device **100** and a learning device **200**. The information processing device **100** is a device that executes an output method. The information processing device **100** outputs a target sound signal by using a learned model. The learned model is generated by the learning device **200**.

(17) The information processing device **100** will be described in a utilization phase. The learning device **200** will be described in a learning phase. First, the utilization phase will be described below.

(18) <Utilization Phase>

(19) FIG. **2** is a diagram showing hardware included in the information processing device in the first embodiment. The information processing device **100** includes a processor **101**, a volatile storage device **102** and a nonvolatile storage device **103**.

(20) The processor **101** controls the whole of the information processing device **100**. The processor **101** is a Central Processing Unit (CPU), a Field Programmable Gate Array (FPGA) or the like, for example. The processor **101** can also be a multiprocessor. Further, the information processing device **100** may include a processing circuitry. The processing circuitry may be either a single circuit or a combined circuit.

(21) The volatile storage device **102** is main storage of the information processing device **100**. The volatile storage device **102** is a Random Access Memory (RAM), for example. The nonvolatile storage device **103** is auxiliary storage of the information processing device **100**. The nonvolatile storage device **103** is a Hard Disk Drive (HDD) or a Solid State Drive (SSD), for example.

(22) Further, a storage area secured by the volatile storage device **102** or the nonvolatile storage device **103** is referred to as a storage unit.

(23) Next, functions included in the information processing device **100** will be described below.

(24) FIG. **3** is a block diagram showing the functions of the information processing device in the first embodiment. The information processing device **100** includes an acquisition unit **120**, a sound feature value extraction unit **130**, an emphasis unit **140**, an estimation unit **150**, a mask feature value extraction unit **160**, a generation unit **170** and a target sound signal output unit **180**.

(25) Part or all of the acquisition unit **120**, the sound feature value extraction unit **130**, the emphasis unit **140**, the estimation unit **150**, the mask feature value extraction unit **160**, the generation unit **170** and the target sound signal output unit **180** may be implemented by a processing circuitry. Further, part or all of the acquisition unit **120**, the sound feature value extraction unit **130**, the emphasis unit **140**, the estimation unit **150**, the mask feature value extraction unit **160**, the generation unit **170** and the target sound signal output unit **180** may be implemented as modules of a program executed by the processor **101**. For example, the program executed by the processor **101** is referred to also as an output program. The output program has been recorded in a record medium, for example.

(26) The storage unit may store sound source position information **111** and a learned model **112**. The sound source position information **111** is position information on a sound source of the target sound. For example, when the target sound is a speech uttered by the target speaker, the sound source position information **111** is position information on the target speaker.

(27) The acquisition unit **120** acquires the sound source position information **111**. For example, the acquisition unit **120** acquires the sound source position information **111** from the storage unit. Here, the sound source position information **111** may be stored in an external device (e.g., cloud server). In the case where the sound source position information **111** is stored in an external device, the acquisition unit **120** acquires the sound source position information **111** from the external device.

(28) The acquisition unit **120** acquires the learned model **112**. For example, the acquisition unit **120** acquires the learned model **112** from the storage unit. Further, for example, the acquisition unit **120** acquires the learned model **112** from the learning device **200**.

(29) The acquisition unit **120** acquires a mixed sound signal. For example, the acquisition unit **120** acquires the mixed sound signal from a microphone array including N microphones (N: integer

greater than or equal to 2). The mixed sound signal is a signal representing a mixed sound including a target sound and a masking sound. The mixed sound signal may be represented as N sound signals. Incidentally, the target sound is a speech uttered by the target speaker, a sound uttered by an animal, or the like, for example. The masking sound is a sound that disturbs the target sound. Further, the mixed sound can include noise. In the following description, the mixed sound is assumed to include the target sound, the masking sound and the noise.

(30) The sound feature value extraction unit **130** extracts a plurality of sound feature values based on the mixed sound signal. For example, the sound feature value extraction unit **130** extracts a time series of power spectra obtained by performing short-time Fourier transform (STFT) on the mixed sound signal as the plurality of sound feature values. Incidentally, the plurality of extracted sound feature values may be represented as N sound feature values.

(31) The emphasis unit **140** emphasizes a sound feature value in a target sound direction among the plurality of sound feature values based on the sound source position information **111**. For example, the emphasis unit **140** emphasizes the sound feature value in the target sound direction by using the plurality of sound feature values, the sound source position information **111** and a Minimum Variance Distortionless Response (MVDR) beam former.

(32) The estimation unit **150** estimates the target sound direction based on the plurality of sound feature values and the sound source position information **111**. Specifically, the estimation unit **150** estimates the target sound direction by using expression (1).

(33) “l” represents time. “k” represents frequency. “x.sub.lk” represents a sound feature value corresponding to a sound signal acquired from a microphone that is the closest to the sound source position of the target sound identified based on the sound source position information **111**.

“x.sub.lk” may be regarded as an STFT spectrum. “a.sub.θ,k” represents a steering vector in a certain angular direction θ. “H” represents conjugate transposition.

$$\theta.\text{sub.lk} = \arg \max_{\theta} |a.\text{sub.}\theta, k.\text{sup.Hx.sub.lk}|.\text{sup.2} \quad (1)$$

(34) The mask feature value extraction unit **160** extracts a mask feature value based on the estimated target sound direction and the plurality of sound feature values. The mask feature value is a feature value in a state in which a feature value in the target sound direction is masked. The process of extracting the mask feature value will be described in detail below. The mask feature value extraction unit **160** generates a directional mask based on the target sound direction. The directional mask is a mask for extracting sound in which the sound of the target sound direction is emphasized. The mask is a matrix of the same size as the sound feature value. When an angular range of the target sound direction is ω, the directional mask M.sub.lk is represented by expression (2)

$$(35) \quad M_{lk} = \begin{cases} 1, & (l_k \in \omega) \\ 0, & (\text{otherwise}) \end{cases} \quad (2)$$

(36) The mask feature value extraction unit **160** extracts the mask feature value by multiplying the plurality of sound feature values by the element-wise product of the mask matrix.

(37) The generation unit **170** generates a sound signal in which the sound of the target sound direction is emphasized (hereinafter referred to as a target sound direction emphasis sound signal) based on the sound feature value emphasized by the emphasis unit **140**. For example, the generation unit **170** generates the target sound direction emphasis sound signal by using the sound feature value emphasized by the emphasis unit **140** and inverse short-time Fourier transform (ISTFT).

(38) The generation unit **170** generates a sound signal in which the sound of the target sound direction is masked (hereinafter referred to as a target sound direction masking sound signal) based on the mask feature value. For example, the generation unit **170** generates the target sound direction masking sound signal by using the mask feature value and inverse short-time Fourier transform.

(39) The target sound direction emphasis sound signal and the target sound direction masking sound signal may be inputted to the learning device **200** as learning signals.

(40) The target sound signal output unit **180** outputs a target sound signal by using the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model **112**. Here, a configuration example of the learned model **112** will be described below.

(41) FIG. **4** is a diagram showing a configuration example of the learned model in the first embodiment. The learned model **112** includes an encoder **112a**, a separator **112b** and a decoder **112c**.

(42) The encoder **112a** estimates a target sound direction emphasis time frequency representation in terms of “ M dimensions $\times$ time” based on the target sound direction emphasis sound signal. Further, the encoder **112a** estimates a target sound direction masking time frequency representation in terms of “ M dimensions $\times$ time” based on the target sound direction masking sound signal. For example, the encoder **112a** may estimate power spectra estimated by means of STFT as the target sound direction emphasis time frequency representation and the target sound direction masking time frequency representation. Further, for example, the encoder **112a** may estimate the target sound direction emphasis time frequency representation and the target sound direction masking time frequency representation by using a one-dimensional convolution operation. When the estimation is performed, the target sound direction emphasis time frequency representation and the target sound direction masking time frequency representation may be either projected onto the same time frequency representation space or projected onto different time frequency representation spaces. Incidentally, this estimation is described in Non-patent Reference 1, for example.

(43) The separator **112b** estimates a mask matrix in terms of “ M dimensions $\times$ time” based on the target sound direction emphasis time frequency representation and the target sound direction masking time frequency representation. Further, when the target sound direction emphasis time frequency representation and the target sound direction masking time frequency representation are inputted to the separator **112b**, the target sound direction emphasis time frequency representation and the target sound direction masking time frequency representation may be connected together in the frequency axis direction. By this, the target sound direction emphasis time frequency representation and the target sound direction masking time frequency representation are transformed into a representation in terms of “ $2M$ dimensions $\times$ time”. The target sound direction emphasis time frequency representation and the target sound direction masking time frequency representation may also be connected together along an axis different from a time axis or a frequency axis. By this, the target sound direction emphasis time frequency representation and the target sound direction masking time frequency representation are transformed into a representation in terms of “ M dimensions $\times$ time $\times 2$ ”. The target sound direction emphasis time frequency representation and the target sound direction masking time frequency representation may be weighted by weights. The weighted target sound direction emphasis time frequency representation and the weighted target sound direction masking time frequency representation may be added together. The weights may be estimated by the learned model **112**.

(44) Incidentally, the separator **112b** is a neural network formed by an input layer, an intermediate layer and an output layer. For example, a method as a combination of a method in the category of Long Short Term Memory (LSTM) and a one-dimensional convolution operation may be used for the propagation between a layer and a layer.

(45) The decoder **112c** multiplies the target sound direction emphasis time frequency representation in terms of “ M dimensions $\times$ time” and the mask matrix in terms of “ M dimensions $\times$ time” together. The decoder **112c** outputs the target sound signal by using the information obtained by the multiplication and a method corresponding to the method used by the encoder **112a**. For example, when the method used by the encoder **112a** is STFT, the decoder **112c** outputs the target sound signal by using the information obtained by the multiplication and ISTFT. Further, for example, when the method used by the encoder **112a** is a one-dimensional convolution operation, the

decoder **112c** outputs the target sound signal by using the information obtained by the multiplication and an inverse one-dimensional convolution operation.

(46) The target sound signal output unit **180** may output the target sound signal to a speaker. Accordingly, the target sound is outputted from the speaker. Incidentally, a drawing of the speaker is left out.

(47) Next, a process executed by the information processing device **100** will be described below by using a flowchart.

(48) FIG. **5** is a flowchart showing an example of the process executed by the information processing device in the first embodiment.

(49) (Step **S11**) The acquisition unit **120** acquires the mixed sound signal.

(50) (Step **S12**) The sound feature value extraction unit **130** extracts the plurality of sound feature values based on the mixed sound signal.

(51) (Step **S13**) The emphasis unit **140** emphasizes the sound feature value in the target sound direction based on the sound source position information **111**.

(52) (Step **S14**) The estimation unit **150** estimates the target sound direction based on the plurality of sound feature values and the sound source position information **111**.

(53) (Step **S15**) The mask feature value extraction unit **160** extracts the mask feature value based on the estimated target sound direction and the plurality of sound feature values.

(54) (Step **S16**) The generation unit **170** generates the target sound direction emphasis sound signal based on the sound feature value emphasized by the emphasis unit **140**. Further, the generation unit **170** generates the target sound direction masking sound signal based on the mask feature value.

(55) (Step **S17**) The target sound signal output unit **180** outputs the target sound signal by using the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model **112**.

(56) Incidentally, the steps **S14** and **S15** may be executed in parallel with the step **S13**. Further, the steps **S14** and **S15** may be executed before the step **S13**.

(57) Next, the learning phase will be described below.

(58) <Learning Phase>

(59) In the learning phase, an example of the generation of the learned model **112** will be described.

(60) FIG. **6** is a block diagram showing functions of the learning device in the first embodiment. The learning device **200** includes a sound data storage unit **211**, an impulse response storage unit **212**, a noise storage unit **213**, an impulse response application unit **220**, a mixing unit **230**, a process execution unit **240** and a learning unit **250**.

(61) Further, the sound data storage unit **211**, the impulse response storage unit **212** and the noise storage unit **213** may be implemented as storage areas secured by a volatile storage device or a nonvolatile storage device included in the learning device **200**.

(62) Part or all of the impulse response application unit **220**, the mixing unit **230**, the process execution unit **240** and the learning unit **250** may be implemented by a processing circuitry included in the learning device **200**. Further, part or all of the impulse response application unit **220**, the mixing unit **230**, the process execution unit **240** and the learning unit **250** may be implemented as modules of a program executed by a processor included in the learning device **200**.

(63) The sound data storage unit **211** stores the target sound signal and a masking sound signal. Incidentally, the masking sound signal is a signal representing the masking sound. The impulse response storage unit **212** stores impulse response data. The noise storage unit **213** stores a noise signal. Incidentally, the noise signal is a signal representing the noise.

(64) The impulse response application unit **220** convolves impulse response data corresponding to the position of the target sound and the position of the masking sound with one target sound signal stored in the sound data storage unit **211** and an arbitrary number of masking sound signals stored in the sound data storage unit **211**.

(65) The mixing unit **230** generates the mixed sound signal based on a sound signal outputted by

the impulse response application unit **220** and the noise signal stored in the noise storage unit **213**. It is also possible to handle the sound signal outputted by the impulse response application unit **220** as the mixed sound signal. The learning device **200** may transmit the mixed sound signal to the information processing device **100**.

(66) The process execution unit **240** generates the target sound direction emphasis sound signal and the target sound direction masking sound signal by executing the steps **S11** to **S16**. In short, the process execution unit **240** generates the learning signals.

(67) The learning unit **250** executes the learning by using the learning signals. In short, the learning unit **250** executes the learning for outputting the target sound signal by using the target sound direction emphasis sound signal and the target sound direction masking sound signal. Incidentally, in the learning, input weight coefficients as parameters of the neural network are determined. In the learning, a loss function described in the Non-patent Reference 1 may be used. Further, in the learning, an error may be calculated by using the sound signal outputted by the impulse response application unit **220** and the loss function. Then, in the learning, an optimization technique such as Adam is used and the input weight coefficient of each layer of the neural network is determined based on error back propagation, for example.

(68) Incidentally, the learning signals may be either the learning signals generated by the process execution unit **240** or the learning signals generated by the information processing device **100**.

(69) Next, a process executed by the learning device **200** will be described below by using a flowchart.

(70) FIG. 7 is a flowchart showing an example of the process executed by the learning device in the first embodiment.

(71) (Step **S21**) The impulse response application unit **220** convolves the impulse response data with the target sound signal and the masking sound signals.

(72) (Step **S22**) The mixing unit **230** generates the mixed sound signal based on the sound signal outputted by the impulse response application unit **220** and the noise signal.

(73) (Step **S23**) The process execution unit **240** generates the learning signals by executing the steps **S11** to **S16**.

(74) (Step **S24**) The learning unit **250** executes the learning by using the learning signals.

(75) Then, the learning device **200** repeats the learning, by which the learned model **112** is generated.

(76) According to the first embodiment, the information processing device **100** outputs the target sound signal by using the learned model **112**. The learned model **112** is a learned model generated by the learning for outputting the target sound signal based on the target sound direction emphasis sound signal and the target sound direction masking sound signal. Specifically, the learned model **112** outputs the target sound signal even when the angle between the target sound direction and the masking sound direction is small by distinguishing between a target sound component having been emphasized or masked and a target sound component not having been emphasized or masked. Therefore, the information processing device **100** is capable of outputting the target sound signal by using the learned model **112** even when the angle between the target sound direction and the masking sound direction is small.

Second Embodiment

(77) Next, a second embodiment will be described below. In the second embodiment, the description will be given mainly of features different from those in the first embodiment. In the second embodiment, the description will be omitted for features in common with the first embodiment.

(78) FIG. 8 is a block diagram showing functions of an information processing device in the second embodiment. The information processing device **100** further includes a selection unit **190**.

(79) Part or the whole of the selection unit **190** may be implemented by a processing circuitry. Further, part or the whole of the selection unit **190** may be implemented as modules of a program

executed by the processor **101**.

(80) The selection unit **190** selects a sound signal in a channel in the target sound direction by using the mixed sound signal and the sound source position information **111**. In other words, the selection unit **190** selects the sound signal in the channel in the target sound direction from the N sound signals based on the sound source position information **111**.

(81) Here, the selected sound signal, the target sound direction emphasis sound signal and the target sound direction masking sound signal may be inputted to the learning device **200** as the learning signals.

(82) The target sound signal output unit **180** outputs the target sound signal by using the selected sound signal, the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model **112**.

(83) Next, processes executed by the encoder **112a**, the separator **112b** and the decoder **112c** included in the learned model **112** will be described below.

(84) The encoder **112a** estimates the target sound direction emphasis time frequency representation in terms of “M dimensions×time” based on the target sound direction emphasis sound signal. Further, the encoder **112a** estimates the target sound direction masking time frequency representation in terms of “M dimensions×time” based on the target sound direction masking sound signal. Furthermore, the encoder **112a** estimates a mixed sound time frequency representation in terms of “M dimensions×time” based on the selected sound signal. For example, the encoder **112a** may estimate power spectra estimated by means of STFT as the target sound direction emphasis time frequency representation, the target sound direction masking time frequency representation and the mixed sound time frequency representation. Further, for example, the encoder **112a** may estimate the target sound direction emphasis time frequency representation, the target sound direction masking time frequency representation and the mixed sound time frequency representation by using a one-dimensional convolution operation. When the estimation is performed, the target sound direction emphasis time frequency representation, the target sound direction masking time frequency representation and the mixed sound time frequency representation may be either projected onto the same time frequency representation space or projected onto different time frequency representation spaces. Incidentally, this estimation is described in the Non-patent Reference 1, for example.

(85) The separator **112b** estimates the mask matrix in terms of “M dimensions×time” based on the target sound direction emphasis time frequency representation, the target sound direction masking time frequency representation and the mixed sound time frequency representation. Further, when the target sound direction emphasis time frequency representation, the target sound direction masking time frequency representation and the mixed sound time frequency representation are inputted to the separator **112b**, the target sound direction emphasis time frequency representation, the target sound direction masking time frequency representation and the mixed sound time frequency representation may be connected together in the frequency axis direction. By this, the target sound direction emphasis time frequency representation, the target sound direction masking time frequency representation and the mixed sound time frequency representation are transformed into a representation in terms of “3M dimensions×time”. The target sound direction emphasis time frequency representation, the target sound direction masking time frequency representation and the mixed sound time frequency representation may also be connected together along an axis different from a time axis or a frequency axis. By this, the target sound direction emphasis time frequency representation, the target sound direction masking time frequency representation and the mixed sound time frequency representation are transformed into a representation in terms of “M dimensions×time×3”. The target sound direction emphasis time frequency representation, the target sound direction masking time frequency representation and the mixed sound time frequency representation may be weighted by weights. The weighted target sound direction emphasis time frequency representation, the weighted target sound direction masking time frequency

representation and the weighted mixed sound time frequency representation may be added together. The weights may be estimated by the learned model **112**.

(86) The process executed by the decoder **112c** is the same as that in the first embodiment.

(87) As above, the target sound signal output unit **180** outputs the target sound signal by using the selected sound signal, the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model **112**.

(88) Next, a process executed by the information processing device **100** will be described below by using a flowchart.

(89) FIG. **9** is a flowchart showing an example of the process executed by the information processing device in the second embodiment. The process of FIG. **9** differs from the process of FIG. **5** in that steps **S11a** and **S17a** are executed. Therefore, the steps **S11a** and **S17a** will be described below with reference to FIG. **9**. Then, the description will be omitted for processing other than the steps **S11a** and **S17a**.

(90) (Step **S11a**) The selection unit **190** selects the sound signal in the channel in the target sound direction by using the mixed sound signal and the sound source position information **111**.

(91) (Step **S17a**) The target sound signal output unit **180** outputs the target sound signal by using the selected sound signal, the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model **112**.

(92) Incidentally, the step **S11a** may be executed at arbitrary timing as long as the step **S11a** is executed before the step **S17a** is executed.

(93) Here, the generation of the learned model **112** will be described below. The learning device **200** executes the learning by using learning signals including the sound signal in the channel in the target sound direction (i.e., mixed sound signal in the target sound direction). For example, the learning signals may be generated by the process execution unit **240**.

(94) The learning device **200** learns the difference between the target sound direction emphasis sound signal and the mixed sound signal in the target sound direction. Further, the learning device **200** learns the difference between the target sound direction masking sound signal and the mixed sound signal in the target sound direction. The learning device **200** learns that a signal in a part where the difference is large is the target sound signal. The learning device **200** executes the learning as above, by which the learned model **112** is generated.

(95) According to the second embodiment, the information processing device **100** is capable of outputting the target sound signal by using the learned model **112** obtained by the learning.

Third Embodiment

(96) Next, a third embodiment will be described below. In the third embodiment, the description will be given mainly of features different from those in the first embodiment. In the third embodiment, the description will be omitted for features in common with the first embodiment.

(97) FIG. **10** is a block diagram showing functions of an information processing device in the third embodiment. The information processing device **100** further includes a reliability calculation unit **191**.

(98) Part or the whole of the reliability calculation unit **191** may be implemented by a processing circuitry. Further, part or the whole of the reliability calculation unit **191** may be implemented as modules of a program executed by the processor **101**.

(99) The reliability calculation unit **191** calculates reliability $F_{sub.i}$ of the mask feature value by a predetermined method. The reliability $F_{sub.i}$ of the mask feature value may be referred to also as reliability $F_{sub.i}$ of the directional mask. The predetermined method is represented by the following expression (3). “ ω ” represents the angular range of the target sound direction. “ θ ” represents an angular range of a direction in which sound occurs.

$$(100) \quad F_i = \frac{\text{Math. } a_{,k} \times_{lk} \text{Math. } ^2}{\text{Math. } a_{,k} \times_{lk} \text{Math. } ^2} \quad (3)$$

(101) The reliability $F_{sub.i}$ is a matrix of the same size as the directional mask. Incidentally, the

reliability F.sub.i may be inputted to the learning device **200**.

(102) The target sound signal output unit **180** outputs the target sound signal by using the reliability F.sub.i, the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model **112**.

(103) Next, processes executed by the encoder **112a**, the separator **112b** and the decoder **112c** included in the learned model **112** will be described below.

(104) The encoder **112a** executes the following process in addition to the process in the first embodiment. The encoder **112a** calculates a time frequency representation FT by multiplying a frame number T and a frequency bin number F of the reliability F.sub.i together. Incidentally, the frequency bin number F is the number of elements of the time frequency representation in the frequency axis direction. The frame number T is a number obtained by dividing the mixed sound signal by a predetermined time.

(105) When the target sound direction emphasis time frequency representation and the time frequency representation FT coincide with each other, the time frequency representation FT is handled in the subsequent processes as the mixed sound time frequency representation in the second embodiment. When the target sound direction emphasis time frequency representation and the time frequency representation FT do not coincide with each other, the encoder **112a** executes a transformation matrix conversion process. Specifically, the encoder **112a** converts the number of elements of the reliability F.sub.i in the frequency axis direction into the number of elements of the target sound direction emphasis time frequency representation in the frequency axis direction.

(106) When the target sound direction emphasis time frequency representation and the time frequency representation FT coincide with each other, the separator **112b** executes the same process as the separator **112b** in the second embodiment.

(107) When the target sound direction emphasis time frequency representation and the time frequency representation FT do not coincide with each other, the separator **112b** integrates the reliability F.sub.i whose number of elements in the frequency axis direction has been converted and the target sound direction emphasis time frequency representation together. For example, the separator **112b** executes the integration by using an attention method described in Non-patent Reference 2. The separator **112b** estimates the mask matrix in terms of “M dimensions×time” based on a target sound direction emphasis time frequency representation obtained by the integration and the target sound direction masking time frequency representation.

(108) The process executed by the decoder **112c** is the same as that in the first embodiment.

(109) As above, the target sound signal output unit **180** outputs the target sound signal by using the reliability F.sub.i, the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model **112**.

(110) Next, a process executed by the information processing device **100** will be described below by using a flowchart.

(111) FIG. **11** is a flowchart showing an example of the process executed by the information processing device in the third embodiment. The process of FIG. **11** differs from the process of FIG. **5** in that steps S15b and S17b are executed. Therefore, the steps S15b and S17b will be described below with reference to FIG. **11**. Then, the description will be omitted for processing other than the steps S15b and S17b.

(112) (Step S15b) The reliability calculation unit **191** calculates the reliability F.sub.i of the mask feature value.

(113) (Step S17b) The target sound signal output unit **180** outputs the target sound signal by using the reliability F.sub.i, the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model **112**.

(114) Here, the generation of the learned model **112** will be described below. When executing the learning, the learning device **200** executes the learning by using the reliability F.sub.i. The learning device **200** may execute the learning by using the reliability F.sub.i acquired from the information

processing device **100**. The learning device **200** may execute the learning by using the reliability $F_{sub.i}$ stored in the volatile storage device or the nonvolatile storage device included in the learning device **200**. The learning device **200** determines how much the target sound direction masking sound signal should be taken into consideration by using the reliability $F_{sub.i}$. The learning device **200** executes learning for making the determination, by which the learned model **112** is generated.

(115) According to the third embodiment, the target sound direction emphasis sound signal and the target sound direction masking sound signal are inputted to the learned model **112**. The target sound direction masking sound signal is generated based on the mask feature value. The learned model **112** determines how much the target sound direction masking sound signal should be taken into consideration by using the reliability $F_{sub.i}$ of the mask feature value. The learned model **112** outputs the target sound signal based on the determination. As above, the information processing device **100** is capable of outputting a more appropriate target sound signal by inputting the reliability $F_{sub.i}$ to the learned model **112**.

Fourth Embodiment

(116) Next, a fourth embodiment will be described below. In the fourth embodiment, the description will be given mainly of features different from those in the first embodiment. In the fourth embodiment, the description will be omitted for features in common with the first embodiment.

(117) FIG. **12** is a block diagram showing functions of an information processing device in the fourth embodiment. The information processing device **100** further includes a noise section detection unit **192**.

(118) Part or the whole of the noise section detection unit **192** may be implemented by a processing circuitry. Further, part or the whole of the noise section detection unit **192** may be implemented as modules of a program executed by the processor **101**.

(119) The noise section detection unit **192** detects a noise section based on the target sound direction emphasis sound signal. For example, the noise section detection unit **192** uses a method described in the Non-patent Reference 2 when detecting the noise section. For example, the noise section detection unit **192** detects speech sections based on the target sound direction emphasis sound signal and thereafter corrects a starting end time of each speech section and a terminal end time of each speech section and thereby identifies the speech sections. The noise section detection unit **192** detects the noise section by excluding the identified speech sections from a section representing the target sound direction emphasis sound signal. Here, the detected noise section may be inputted to the learning device **200**.

(120) The target sound signal output unit **180** outputs the target sound signal by using the detected noise section, the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model **112**.

(121) Next, processes executed by the encoder **112a**, the separator **112b** and the decoder **112c** included in the learned model **112** will be described below.

(122) The encoder **112a** executes the following process in addition to the process in the first embodiment. The encoder **112a** estimates a non-target sound time frequency representation in terms of “M dimensions×time” based on a signal corresponding to the noise section of the target sound direction emphasis sound signal. For example, the encoder **112a** may estimate a power spectrum estimated by means of STFT as the non-target sound time frequency representation. Further, for example, the encoder **112a** may estimate the non-target sound time frequency representation by using a one-dimensional convolution operation. When the estimation is performed, the non-target sound time frequency representation may be either projected onto the same time frequency representation space or projected onto a different time frequency representation space. Incidentally, this estimation is described in the Non-patent Reference 1, for example.

(123) The separator **112b** integrates the non-target sound time frequency representation and the target sound direction emphasis time frequency representation together. For example, the separator **112b** executes the integration by using the attention method described in the Non-patent Reference 2. The separator **112b** estimates the mask matrix in terms of “M dimensions×time” based on a target sound direction emphasis time frequency representation obtained by the integration and the target sound direction masking time frequency representation.

(124) Incidentally, the separator **112b** is capable of estimating a tendency of the noise based on the non-target sound time frequency representation, for example.

(125) The process executed by the decoder **112c** is the same as that in the first embodiment.

(126) Next, a process executed by the information processing device **100** will be described below by using a flowchart.

(127) FIG. **13** is a flowchart showing an example of the process executed by the information processing device in the fourth embodiment. The process of FIG. **13** differs from the process of FIG. **5** in that steps **S16c** and **S17c** are executed. Therefore, the steps **S16c** and **S17c** will be described below with reference to FIG. **13**. Then, the description will be omitted for processing other than the steps **S16c** and **S17c**.

(128) (Step **S16c**) The noise section detection unit **192** detects the noise section as section representing the noise based on the target sound direction emphasis sound signal.

(129) (Step **S17c**) The target sound signal output unit **180** outputs the target sound signal by using the noise section, the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model **112**.

(130) Here, the generation of the learned model **112** will be described below. When executing the learning, the learning device **200** executes the learning by using the noise section. The learning device **200** may execute the learning by using the noise section acquired from the information processing device **100**. The learning device **200** may execute the learning by using the noise section detected by the process execution unit **240**. The learning device **200** learns the tendency of the noise based on the noise section. The learning device **200** executes the learning for outputting the target sound signal based on the target sound direction emphasis sound signal and the target sound direction masking sound signal and in consideration of the tendency of the noise. The learning device **200** executes the learning as above, by which the learned model **112** is generated.

(131) According to the fourth embodiment, the noise section is inputted to the learned model **112**. The learned model **112** estimates the tendency of the noise included in the target sound direction emphasis sound signal and the target sound direction masking sound signal based on the noise section. The learned model **112** outputs the target sound signal based on the target sound direction emphasis sound signal and the target sound direction masking sound signal and in consideration of the tendency of the noise. Since the information processing device **100** outputs the target sound signal in consideration of the tendency of the noise, the information processing device **100** is capable of outputting a more appropriate target sound signal.

(132) Features in the embodiments described above can be appropriately combined with each other.

DESCRIPTION OF REFERENCE CHARACTERS

(133) **100**: information processing device, **101**: processor, **102**: volatile storage device, **103**: nonvolatile storage device, **111**: sound source position information, **112**: learned model, **120**: acquisition unit, **130**: sound feature value extraction unit, **140**: emphasis unit, **150**: estimation unit, **160**: mask feature value extraction unit, **170**: generation unit, **180**: target sound signal output unit, **190**: selection unit, **191**: reliability calculation unit, **192**: noise section detection unit, **200**: learning device, **211**: sound data storage unit, **212**: impulse response storage unit, **213**: noise storage unit, **220**: impulse response application unit, **230**: mixing unit, **240**: process execution unit, **250**: learning unit.

Claims

1. An information processing device comprising: acquiring circuitry to acquire sound source position information as position information on a sound source of a target sound, a mixed sound signal as a signal representing a mixed sound including the target sound and a masking sound, and a learned model; sound feature value extracting circuitry to extract a plurality of sound feature values based on the mixed sound signal, the plurality of sound feature values being a time series of power spectra obtained by performing short-time Fourier transform (STFT) on the mixed sound signal; emphasizing circuitry to emphasize a sound feature value in a target sound direction as a direction of the target sound among the plurality of sound feature values based on the sound source position information thereby amplifying a particular part of the power spectra in the time series relative to other parts, wherein the target sound direction is the direction from which the target sound is emitted from the sound source of the target sound; estimating circuitry to estimate the target sound direction based on the plurality of sound feature values and the sound source position information; mask feature value extracting circuitry to extract a mask feature value, as a feature value in a state in which a feature value in the target sound direction is masked, based on the estimated target sound direction and the plurality of sound feature values; generating circuitry to generate a target sound direction emphasis sound signal, as a sound signal in which the sound of the target sound direction is emphasized, based on the emphasized sound feature value and generate a target sound direction masking sound signal, as a sound signal in which the sound of the target sound direction is masked, based on the mask feature value; and target sound signal outputting circuitry to output a target sound signal as a signal representing the target sound by using the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model.
2. The information processing device according to claim 1, further comprising selecting circuitry to select a sound signal in a channel in the target sound direction by using the mixed sound signal and the sound source position information, wherein the target sound signal outputting circuitry outputs the target sound signal by using the selected sound signal, the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model.
3. The information processing device according to claim 1, further comprising reliability calculating circuitry to calculate reliability of the mask feature value by a predetermined method, wherein the target sound signal outputting circuitry outputs the target sound signal by using the reliability, the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model.
4. The information processing device according to claim 1, wherein the mixed sound includes noise.
5. The information processing device according to claim 4, further comprising noise section detecting circuitry to detect a noise section as a section representing the noise based on the target sound direction emphasis sound signal, wherein the target sound signal outputting circuitry outputs the target sound signal by using the noise section, the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model.
6. An output method performed by an information processing device, the output method comprising: acquiring sound source position information as position information on a sound source of a target sound, a mixed sound signal as a signal representing a mixed sound including the target sound and a masking sound, and a learned model; extracting a plurality of sound feature values based on the mixed sound signal, the plurality of sound feature values being a time series of power spectra obtained by performing short-time Fourier transform (STFT) on the mixed sound signal; emphasizing a sound feature value in a target sound direction as a direction of the target sound among the plurality of sound feature values based on the sound source position information thereby

amplifying a particular part of the power spectra in the time series relative to other parts, wherein the target sound direction is the direction from which the target sound is emitted from the sound source of the target sound; estimating the target sound direction based on the plurality of sound feature values and the sound source position information; extracting a mask feature value, as a feature value in a state in which a feature value in the target sound direction is masked, based on the estimated target sound direction and the plurality of sound feature values; generating a target sound direction emphasis sound signal, as a sound signal in which the sound of the target sound direction is emphasized, based on the emphasized sound feature value, generating a target sound direction masking sound signal, as a sound signal in which the sound of the target sound direction is masked, based on the mask feature value; and outputting a target sound signal as a signal representing the target sound by using the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model.

7. An information processing device comprising: a processor to execute a program; and a memory to store the program which, when executed by the processor, performs processes of, acquiring sound source position information as position information on a sound source of a target sound, a mixed sound signal as a signal representing a mixed sound including the target sound and a masking sound, and a learned model, extracting a plurality of sound feature values based on the mixed sound signal, the plurality of sound feature values being a time series of power spectra obtained by performing short-time Fourier transform (STFT) on the mixed sound signal, emphasizing a sound feature value in a target sound direction as a direction of the target sound among the plurality of sound feature values based on the sound source position information thereby amplifying a particular part of the power spectra in the time series relative to other parts, wherein the target sound direction is the direction from which the target sound is emitted from the sound source of the target sound, estimating the target sound direction based on the plurality of sound feature values and the sound source position information, extracting a mask feature value, as a feature value in a state in which a feature value in the target sound direction is masked, based on the estimated target sound direction and the plurality of sound feature values, generating a target sound direction emphasis sound signal, as a sound signal in which the sound of the target sound direction is emphasized, based on the emphasized sound feature value, generating a target sound direction masking sound signal, as a sound signal in which the sound of the target sound direction is masked, based on the mask feature value, and outputting a target sound signal as a signal representing the target sound by using the target sound direction emphasis sound signal, the target sound direction masking sound signal and the learned model.
